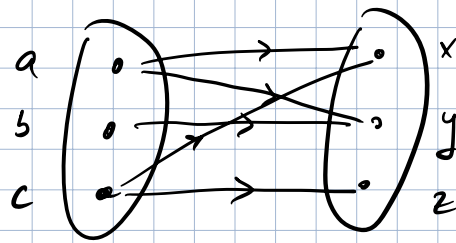


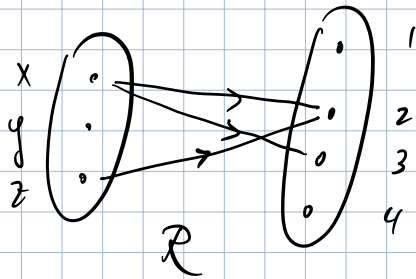
Adjacency matrices and composition:

from
to $\begin{pmatrix} \\ \\ \end{pmatrix}$



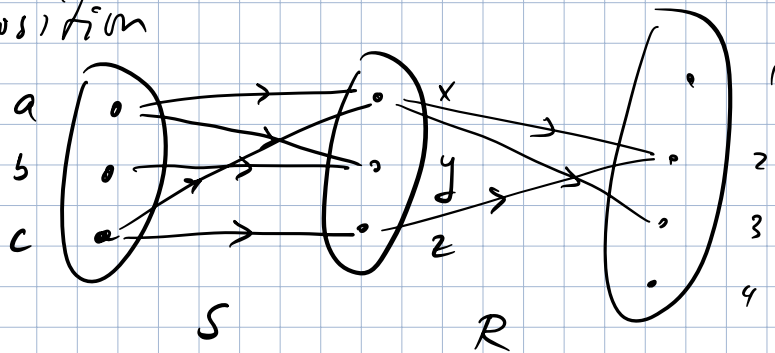
$$\begin{matrix} & a & b & c \\ \begin{matrix} x \\ y \\ z \end{matrix} & \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{matrix}$$

S relation



$$\begin{matrix} & x & y & z \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

Composition



composition
 $R \circ S$
R after S

Multiply the adjacency matrices accordingly

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{matrix} & a & b & c \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 2 \\ 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} \end{matrix}$$

$c \text{ R.o.S } 2$ is 'counted twice'.